HOORVIA - SQL SYNTAX CHART: PRINTABLE QUERY REFERENCE

Quick Reference Cheat Sheet · A4 Printable

Tech & Software · Sheet #1

SQL Clause / Command	Standard ANSI SQL Syntax	Relational Purpose	Performance / Indexing Tip
SELECT	SELECT col1, col2 FROM t	Picks the columns to return	Use explicit columns, never *
WHERE	SELECT * FROM t WHERE cond	Filters the rows	Filter early to cut rows
INSERT	INSERT INTO t (a, b) VALUES (1, 2)	Adds new rows	Batch inserts for speed
UPDATE	UPDATE t SET a = 1 WHERE id = 5	Changes existing rows	Always include WHERE
DELETE	DELETE FROM t WHERE id = 5	Removes rows	Always include WHERE
ORDER BY	SELECT * FROM t ORDER BY col DESC	Sorts the results	Index the sort column
LIMIT	SELECT * FROM t LIMIT 10	Caps the rows returned	Use with ORDER BY
INNER JOIN	SELECT * FROM a INNER JOIN b ON a.id = b.a_id	Returns records with matching keys in both tables	Ensure foreign keys have B-tree indexes on joining columns
LEFT OUTER JOIN	SELECT * FROM a LEFT JOIN b ON a.id = b.a_id	Returns all records from left table and matched from right	Use to identify orphan rows with WHERE b.id IS NULL
GROUP BY / HAVING	SELECT dept, COUNT(*) FROM emp GROUP BY dept HAVING COUNT(*) > 5	Aggregates rows by common values with post-group filtering	Filter early with WHERE before GROUP BY to reduce hash buffer size
Window Function	ROW_NUMBER() OVER (PARTITION BY dept ORDER BY salary DESC)	Calculates ranked metric across row subset without collapsing rows	Optimal for top-N ranking per partition without subqueries
Index Creation	CREATE INDEX idx_user_email ON users(email);	Accelerates search queries and equality filters	Index frequently filtered/joined columns; avoid indexing low-cardinality columns

PRO TIP: Write the WHERE first: An UPDATE or DELETE without a WHERE changes every row, and there is no undo. Write the WHERE, check it with a SELECT, then run the change.